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Revisiting Group Differences in High-Dimensional Choices: Method and Application to Congressional Speech

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ABSTRACT

Gentzkow, Shapiro, and Taddy, *Econometrica* 87, no. 4, 2019 (henceforth GST), use a supervised text-based regression model to assess changes in partisanship in US congressional speech over time. Their estimates imply that partisanship is far greater in recent years than in the past and that it increased sharply in the early 1990s. The paper at hand provides a replication in the wide sense of GST by complementing their analysis in three ways. First, we propose an alternative unsupervised language model, which combines ideas of topic models and ideal point models, to analyze the change in partisanship over time. We apply this model to the Senate speech data used in GST ranging from 1981 to 2017. Using our model, we replicate their results on the specific evolution of partisanship. Second, our model provides additional insights such as the data-driven estimation of evolution of topical contents over time. Third, we identify key phrases of partisanship on topic level.

1 | Introduction

Estimating political positions—so called ideal points—of political key actors has a long tradition in political science (see, e.g., Poole and Rosenthal 1985; McCarty et al. 1997). Gentzkow et al. (2019b) (henceforth GST) contribute to this literature on an aggregate level by proposing a text-based regression model to study to what extent partisanship between political parties, estimated from differences in word usage in speeches, evolves over time. They use transcripts of the speeches given in the US Congress between 1873 and 2016 (Congress Sessions 43–114). Their main result is that partisanship in Congress speeches is a relatively recent phenomenon. Its inflection point coincides precisely with innovations in political persuasion in the 1990s formulated under a platform called the *Contract with America*. Methodologically, GST build on Taddy (2013) to learn a predictive model for party given word usage. They also investigate differences in partisanship according to 22 topics that they identified

manually based on domain knowledge by assigning key terms to characterize them.

The paper at hand presents the results obtained with an alternative statistical model to assess the changes in partisanship over time. Our model is a time-varying (TV) extension of the text-based ideal point model (TBIP; Vafa et al. 2020) (henceforth TV-TBIP). TV-TBIP combines the class of topic models (see Blei et al. 2003) and text-based ideal point models (see, e.g., Slapin and Proksch 2008; Lauderdale and Herzog 2016). In contrast to the regression-based approach pursued in GST, TV-TBIP is an unsupervised modeling approach where topics are extracted in a data-driven way allowing their evolution over time.

We apply our model to speeches given in the US Senate in sessions 97–114, that is, between 1981 and 2017, making use of the speech text data provided as supplementary material by GST.¹

The authors want to thank Keyon Vafa for many helpful comments and feedback. This research was supported by the OeNB Jubiläumsfonds (project number: 18718).

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Our model estimates for each speaker a session-specific ideological position on a one-dimensional latent scale. This latent scale is shown to primarily capture the party differences. Ideal points capturing the ideological positions thus allow obtaining an aggregate measure of partisanship for each session. In line with GST, we find that Congress Session 105 is the first session where a clear separation between the party-specific distributions of the ideal points is discernible. In contrast to GST, our model provides partisanship estimates on the speaker level, and these individual ideological positions allow to infer which speakers are at the extremes of the latent scale or rather centrally located and to track changes of their ideological positions over time. TV-TBIP extends the approach taken in GST by estimating the topics in a data-driven way. In addition, TV-TBIP also allows term compositions of topics to change over time to account for newly emerging or subduing subthemes within a topic. This is particularly important when analyzing more than 40 years of speeches given in the US Senate.

The paper is organized as follows: Section 2 introduces the time-varying text-based ideal point (TV-TBIP). Section 3 discusses the results and insights gained from fitting the TV-TBIP model and contrasts to them to those in GST. Finally, Section 4 concludes.

2 | The Time-Varying Text-Based Ideal Point Model

The TBIP model proposed in Vafa et al. (2020) assumes that documents are composed of a fixed set of topics (latent themes) where ideological differences of the authors influence the term compositions of topics. Considering a corpus of speeches covering an extensive time period of about 40 years, we extend the TBIP model to a time-varying version which allows the term compositions of the topics as well as the individual ideological positions to adapt over time.

2.1 | Model Specification

The available data for each session or time point t consist of the text of the speeches given as well as the information on the speaker giving the speech. Based on the bag-of-words assumption, the speech data are summarized in a document-term matrix c^t . This matrix contains the frequency counts of the terms in each speech at time point t with the number of rows corresponding to the number of speeches and the number of columns to the vocabulary size. The speaker information is combined in the vector s^t with length equal to the number of speeches.

The parameters and latent variables inferred in the TV-TBIP model for each time point t consist of document-specific topic prevalences θ^t , topic-specific term prevalences for neutral speakers β^t , topic-specific polarity scores for the terms η^t , and individual ideal points (positions) of the speakers x^t . The matrix of topic distributions at time point t , θ^t , has dimension number of speeches times number of topics. The matrix β^t is of dimension number of topics times the size of the vocabulary. η^t has the same dimension as β^t , and the vector of ideal points x^t has the length of the number of speakers at time t .

The TV-TBIP model assumes that the observed frequencies of the terms in the speeches are generated independently from a Poisson distribution; that is, the frequency count c_{iv}^t at time point t for speech i and term v is generated by

$$c_{iv}^t \sim \text{Pois} \left(\sum_{k=1}^K \theta_{ik}^t \beta_{kv}^t \exp \left\{ x_{is}^t \eta_{kv}^t \right\} \right). \quad (1)$$

The Poisson rates are derived as linear combinations of contributions from the K topics where θ_{ik}^t is the topic prevalence of topic k in speech i at time point t and β_{kv}^t is the term prevalence of term v in topic k at time point t of a neutral speaker; that is, these term prevalences characterize “neutral topics.”

For each speaker s , the ideological position is encapsulated in the ideal point x_s^t . For a neutral speaker with $x_s^t = 0$, the model reduces to the simple Poisson factorization topic model (Canny 2004). With ideological positions departing from the neutral one, the Poisson rates of the terms for each of the topics also differ from the neutral one. If the ideal point x_s^t and the prevalence modification η_{kv}^t at time point t for term v and topic k have the same sign, a speaker will more often use this term when talking about this topic compared to a neutral speaker. If they have opposite signs, usage of that term for this topic is decreased compared to someone who is neutral. The TV-TBIP model assumes that the polarity of a speaker captured by the ideal points influences their word choice for a specific topic; that is, the term prevalences of a specific topic differ between a negative, a neutral, and a positive speaker. By contrast, the model assumes that ideal points do not impact on the topic prevalences; that is, a negative, a neutral, and a positive speaker talk to a similar extent about each of the topics.

2.2 | Model Estimation

We estimate the model within a Bayesian framework. The parameters and latent variables are assumed to be independent and identically distributed with the same prior settings used for each time point, speech, term, and topic as well as speaker.

The priors for the topic and term prevalences θ_{ik}^t and β_{kv}^t are assumed to follow independent Gamma distributions with potentially different hyperparameters for θ_{ik}^t and β_{kv}^t but identical Gamma distributions for all sessions t , speeches i , terms v , and topics k :

$$\theta_{ik}^t \sim \text{Gamma}(\alpha_1, \alpha_2), \quad \beta_{kv}^t \sim \text{Gamma}(\gamma_1, \gamma_2).$$

Careful selection of the parameters of the Gamma distributions is required in order to induce a suitable sparsity in the topic prevalences of the speeches and in the term prevalences of the topics. Sparsity implies that each speech consists of a small number of topics and that each topic is characterized by a few terms with high prevalence values. The choice of these hyperparameters impacts on the specific solution obtained. Here, we follow Vafa et al. (2020) and set $\alpha_1 = \alpha_2 = 0.3$ and $\gamma_1 = \gamma_2 = 0.3$. To facilitate interpretability of the topics obtained, such a sparsity characteristic is desirable. In addition, this alleviates the issue of non-identifiability which topic models are known to suffer from and which we would expect to also be an issue for TV-TBIP.

For non-identifiable topic models, Ke et al. (2024) highlight the importance of selecting suitable priors which support to prefer a specific parameterization among those inducing the same likelihood.

The topic polarity scores and ideal points η_{kv}^t and x_s^t can take arbitrary values, negative and positive. To reflect this unrestricted support, standard normal distributions are imposed as their priors. This means that for all sessions t , terms v , topics k , and speakers s , we assume independent distributions given by

$$\eta_{kv}^t \sim \text{Normal}(0, 1), \quad x_s^t \sim \text{Normal}(0, 1).$$

Restricting the mean values to zero implies that an average speaker in the latent dimension is viewed as a neutral speaker and the average intensities across the latent dimension correspond to the neutral topic intensities. Setting the variance to one improves identifiability of the model, as this fixes the possible within-topic variability due to changes in the polarity of the speaker.

The parameters and latent variables are estimated for each session separately by approximating their posterior distribution given the priors and the observed data. The posterior of interest $p(\theta^t, \beta^t, \eta^t, x^t | c^t, s^t)$ is not available in closed-form, and it is also computationally intractable to directly obtain estimates from the posterior. One thus usually resorts to approximation using variational inference (Blei et al. 2017). Using variational inference, the parameters and latent variables are estimated based on an approximation of the posterior obtained by minimizing the Kullback–Leibler divergence between the posterior and a variational distribution. Point estimates are obtained by considering the posterior means implied for the parameters and latent variables by the fitted variational distributions.

This model specification and estimation scheme enables the derivation of session-specific parameter and latent variable estimates that capture differences and adaptations over time. However, we are also interested in linking topics as well as the latent space over time. This is particularly relevant due to the identifiability issues present for the TBIP model, that is, neither the topics nor the latent space are identifiable. To obtain a time-varying version of the TBIP model, where the parameters are congruent across time, we make use of a step-wise initialization approach across sessions where the estimates obtained in the previous session are used as initial values in the next session as far as possible when using a general purpose optimizer to minimize the Kullback–Leibler divergence which requires some initial values to be provided anyway.

In the first session, no previous results are available, and non-negative matrix factorization (NMF; Lee and Seung 1999) is used to initialize the variational parameters for θ^1 and β^1 . The input to the NMF algorithm is the document-term matrix of the first session (i.e., c^1), which is approximated as the product of the topical prevalence matrix θ^1 and topical content matrix β^1 , that is, $c^1 \simeq \theta^1 \beta^1$. These two matrices obtained with NMF are first ensured to only contain positive entries and then the logarithm is taken. The resulting values are used to initialize the means of the log-normal distributions used as variational distributions for these parameters.

For each subsequent session with time index $t \geq 2$, the estimated posterior means of the term prevalence vectors of the topics β^{t-1} are used to initialize the mean parameters of the corresponding variational parameters in the TV-TBIP model. These estimated prevalences are also used as input in the NMF algorithm together with the document-term matrix c^t such that only the initial values for the topic distributions θ^t of the speeches are determined in this step; that is, $c^t \simeq \theta^t \beta^{t-1}$ is used for initialization. In addition, the estimated values of η^{t-1} are put forward as the means of the Gaussian variational family of η^t . Specifying the initial values in this way ensures that the term prevalences of topics and the topic-specific polarity scores are aligned across sessions preventing, for example, label switching of topics and reversing of the latent space.² For more details on the model estimation, we refer to Appendix S1.

3 | Forty Years of Political Discourse in the US Senate

We use the Stanford University Social Science Data Collection database for our analysis (Gentzkow et al. 2018). This database was also used in GST and provides pre-processed text data on the speech level from the United States Congressional Record, covering Congress sessions 97–114 (1981–2017). We use all speeches given in the US Senate and implement pre-processing steps similar to GST and Vafa et al. (2020) to obtain session-specific document-term matrices, using bigrams as tokens and retaining only those used by at least 10 speakers per session. We also only retain speakers with at least 24 speeches per session. Our model is fitted using 614,613 speeches given by 355 unique speakers, assuming 25 topics per session. This is in line with GST who manually identified 22 substantive topics based on their knowledge of the database. More details on data pre-processing and implementation of the estimation are given in Appendix S1.

3.1 | Average Partisanship

We inspect the session- and party-specific distributions of the estimated ideal points to assess if the latent scale in fact differentiates between Democrat and Republican speakers for each session. Figure 1 (left) summarizes these ideal point distributions based on box-plots. Note that the ideal points are a priori assumed to be standard normally distributed with the mean zero indicating a neutral speaker. The values of the estimated ideal points may thus be interpreted as reflecting the degree of partisanship of the speakers with the distance of the ideal point from zero indicating the deviation of the speaker from a neutral speaker in the unit of standard deviations. For each session, a pair of box-plots is displayed for the two parties. This provides a means to compare the party-specific locations based on the medians and assess the overlap based on the boxes. Clearly, the boxes overlap strongly in the first sessions, while boxes do not overlap after session 105.

To study the evolvement of average partisanship across Congress sessions, we determine an estimate of the average partisanship between the parties over time by aggregating the point estimates

TABLE 1 | Most frequent terms for each topic for session 97 (left) and session 114 (right).

#	Topic	Session 97 (from January 3, 1981, to January 3, 1983)	Session 114 (from January 3, 2015, to January 3, 2017)
1	United States	united states, foreign policy, international trade, foreign relations, acid rain (0.315) (0.008) (0.007) (0.006) (0.005)	united states, puerto rico, trade agreements, trade promotion, promotion authority (0.389) (0.022) (0.021) (0.014)
2	Export/import	private sector, small businesses, tax credit, internal revenue, clean air (0.03) (0.021) (0.019) (0.016) (0.011)	small businesses, american workers, exportimport bank, exim bank, economic growth (0.049) (0.02) (0.026) (0.025) (0.019)
3	Commemoration	legal services, services corporation, legal assistance, older americans, human resources (0.071) (0.032) (0.007) (0.007) (0.006)	medical research, state university, high school, passed away, national institutes (0.024) (0.02) (0.018) (0.018) (0.013)
4	Supreme court	supreme court, federal courts, public schools, federal court, school prayer (0.095) (0.03) (0.015) (0.015) (0.014)	supreme court, attorney general, justice scalia, district court, judicial nominees (0.146) (0.03) (0.017) (0.015) (0.013)
5	Finances	interest rates, high interest, federal reserve, interest rate, reserve board (0.13) (0.048) (0.035) (0.021) (0.016)	wall street, federal reserve, protection act, financial crisis, monetary policy (0.061) (0.02) (0.012) (0.011) (0.008)
6	Transport	natural gas, energy assistance, gas prices, policy act, lowincome energy (0.077) (0.015) (0.011) (0.01) (0.008)	home state, important issue, critically important, transportation system, surface transportation (0.086) (0.021) (0.016) (0.01) (0.012)
7	Law enforcement	law enforcement, criminal justice, violent crime, coast guard, attorney general (0.045) (0.018) (0.018) (0.017) (0.014)	law enforcement, human trafficking, mental health, gun violence, criminal justice (0.082) (0.026) (0.021) (0.013) (0.014)
8	Middle East	saudi arabia, middle east, foreign policy, persian gulf, foreign relations (0.078) (0.068) (0.026) (0.018) (0.015)	national security, middle east, united states, al qaeda, islamic state (0.143) (0.034) (0.025) (0.003) (0.016)
9	Budget	balanced budget, federal budget, federal spending, budget process, debt limit (0.071) (0.032) (0.029) (0.022) (0.017)	appropriations process, balanced budget, control act, budget act, budget control (0.019) (0.018) (0.017) (0.013) (0.014)
10	Education	budget authority, budget act, supplemental appropriations, block grant, loan program (0.033) (0.016) (0.011) (0.01) (0.01)	high school, left behind, child left, higher education, school districts (0.057) (0.03) (0.025) (0.023) (0.015)
11	Climate change	nuclear waste, spent fuel, nuclear power, clinch river, public health (0.045) (0.024) (0.022) (0.018) (0.015)	climate change, public health, clean water, zika virus, fossil fuel (0.074) (0.045) (0.03) (0.021) (0.016)
12	Taxes	tax cut, tax cuts, tax increase, economic recovery, tax rates (0.061) (0.029) (0.015) (0.014) (0.013)	middle class, tax credit, tax code, republican budget, tax reform (0.057) (0.028) (0.025) (0.02) (0.018)
13	Civil rights	voting rights, rights act, civil rights, justice department, attorney general (0.06) (0.048) (0.036) (0.02) (0.015)	civil rights, voting rights, equal pay, rights act, fair housing (0.046) (0.022) (0.021) (0.018) (0.014)

(Continues)

TABLE 1 | (Continued)

#	Topic	Session 97 (from January 3, 1981, to January 3, 1983)	Session 114 (from January 3, 2015, to January 3, 2017)
14	Federal government	federal government, local governments, public law, federal funds, federal agencies (0.109) (0.022) (0.009) (0.008)	federal government, private sector, taxpayer dollars, federal agencies, local governments (0.157) (0.037) (0.026)
15	Health care	health care, veterans administration, older americans, medical care, senior citizens (0.057) (0.016) (0.015) (0.014) (0.011)	health care, affordable care, care act, health insurance, planned parenthood (0.161) (0.04) (0.039) (0.038)
16	Rhetoric	peace corps, foreign relations, distinguished colleague, intelligence activities, equally divided (0.081) (0.022) (0.009) (0.008) (0.005)	next week, equally divided, en bloc, important legislation, final passage (0.046) (0.045) (0.035) (0.025) (0.017)
17	Wars	soviet union, world war, foreign policy, war ii, chemical weapons (0.136) (0.026) (0.02) (0.018) (0.013)	world war, war ii, puerto rico, vietnam war, loved ones (0.061) (0.051) (0.045) (0.014) (0.013)
18	Nuclear arms	nuclear weapons, arms control, nuclear war, foreign relations, nuclear arms (0.048) (0.045) (0.025) (0.022) (0.016)	united states, nuclear weapons, nuclear weapon, nuclear program, middle east (0.037) (0.035) (0.031) (0.02)
19	Social security	social security, security system, minimum benefit, trust fund, security benefits (0.22) (0.028) (0.02) (0.02) (0.018)	social security, trust fund, highway trust, older americans, security administration (0.142) (0.056) (0.023) (0.011)
20	Army	armed services, air force, armed forces, basing mode, national guard (0.059) (0.05) (0.022) (0.02) (0.017)	air force, armed services, defense authorization, national defense, national guard (0.051) (0.048) (0.045) (0.042) (0.023)
21	Natural resources	oil companies, natural resources, windfall profit, crude oil, profit tax (0.025) (0.022) (0.017) (0.017) (0.016)	natural resources, keystone xl, natural gas, energy efficiency, water conservation (0.021) (0.02) (0.019) (0.017)
22	Human rights	human rights, united nations, genocide convention, foreign relations, state department (0.08) (0.03) (0.027) (0.02) (0.019)	human rights, state department, foreign relations, religious freedom, united nations (0.056) (0.041) (0.032) (0.023) (0.019)
23	Security	national security, inspector general, defense department, national defense, defense spending (0.051) (0.033) (0.023) (0.021) (0.014)	homeland security, cyber security, usa freedom, freedom act, inspector general (0.108) (0.034) (0.017) (0.015)
24	Puerto Rico	food stamp, school lunch, food stamps, puerto rico, stamp program (0.04) (0.021) (0.02) (0.019) (0.019)	economic development, local communities, puerto rico, foster care, much needed (0.016) (0.014) (0.011) (0.01)
25	Party rhetoric	executive branch, legislative veto, regulatory reform, federal trade, trade commission (0.041) (0.032) (0.024) (0.018) (0.017)	bipartisan basis, good friend, bipartisan way, democratic colleagues, common ground (0.018) (0.014) (0.011) (0.011)

Note: Values in parentheses denote appearance rates.

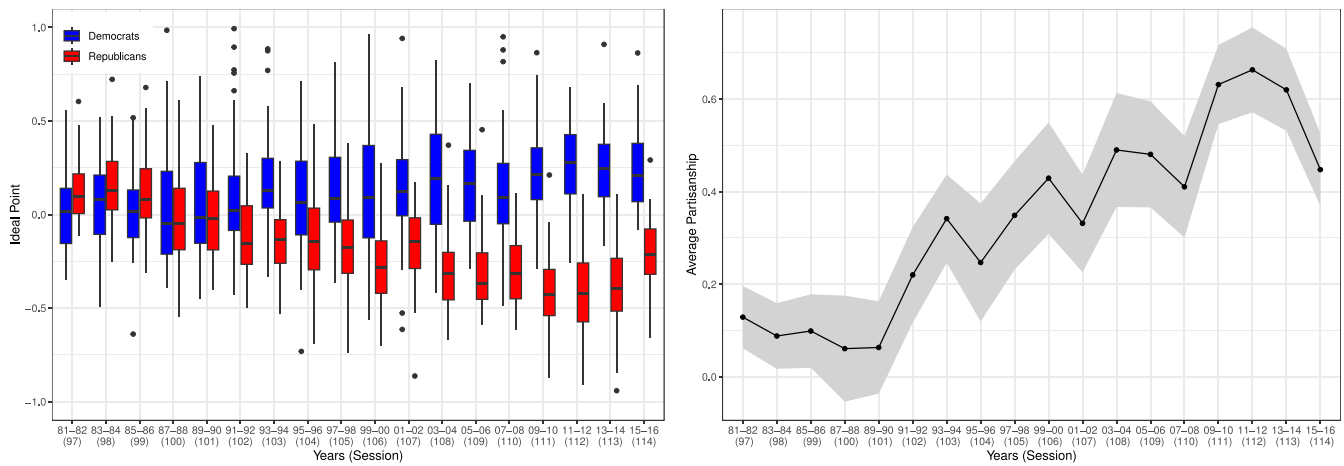


FIGURE 1 | Party- and session-specific ideal point distributions represented as box-plots (left). Estimated average partisanship over the years together with approximate pointwise 95% confidence intervals (right).

of the ideological positions $\hat{x}_s^t$ of the speakers across parties. For each session, we calculate the mean of the estimated ideal points separately for the members of each party and take the difference. That is, the average session-specific partisanship for session t is determined using

$$\bar{\pi}^t = \left| \frac{1}{N_R^t} \sum_{s \in I_R^t} \hat{x}_s^t - \frac{1}{N_D^t} \sum_{s \in I_D^t} \hat{x}_s^t \right|, \quad (2)$$

where N_R^t and N_D^t are the number of Republican and Democrat speakers with ideal point estimates available for session t and I_R^t and I_D^t denote the index sets for those speaker groups. This approach neglects the uncertainty of the Bayesian point estimates and treats the estimates of these latent variables as if they were observed. Alternatively, a fully Bayesian approach could also be pursued.

Figure 1 (right) depicts the evolution of these average partisanship estimates over time and corroborates the key results of GST. The gray shaded area represents the approximate pointwise 95% confidence intervals. These confidence intervals are obtained using $\bar{\pi}^t \pm 1.96 \times \sqrt{\frac{(\hat{\sigma}_D^t)^2}{N_D^t} + \frac{(\hat{\sigma}_R^t)^2}{N_R^t}}$, where $\hat{\sigma}_D^t$ and $\hat{\sigma}_R^t$ are the sample standard deviations of the estimated point estimates for the ideal points for each party. In line with GST, the TV-TBIP model does not detect any noteworthy partisanship in the speeches given in the US Senate before the 1990s. However, at the beginning of the 1990s, the estimated average partisanship between Democrats and Republicans increases rapidly and peaks around 2010. This pattern of evolution of partisanship closely follow the results obtained using the supervised model proposed in GST (see Gentzkow et al. 2019a, Figure 2 on p. 1321). Appendix S1 provides results of a simulation study where the same estimation method is used for analysis. New document-term matrices for each session are generated based on the estimated posterior mean estimates and varying the distribution of ideal points across parties and sessions to induce no, fixed, and increasing party differences as well as re-using the estimated ideal points for the speakers across sessions. These results confirm the suitability of the proposed approach to infer the evolution of partisanship across sessions.

In line with GST, we also validate our aggregate measure of partisanship by comparing it with the first dimension of the DW-Nominate scores estimated from voting data (see, e.g., Poole and Rosenthal 1985). In particular, we take the difference between the average DW-Nominate scores of Republicans and Democrats. We find a correlation of 0.77 indicating that our text-based average partisanship measure captures essentially the same effect over time as the vote-based DW-Nominate scores. Appendix S1 provides more details on the timely evolution of ideological positions on speaker level.

Our model tracks the timely evolution of ideological positions on individual speaker level. Due to the pronounced partisanship from session 105 onwards, we focus in the following analysis on results from session 105 and later (see also Appendix S1 for more details). We determine the average ideal point score for each speaker across all sessions they were in the Senate during this time period. These results indicate that the most liberal Democrats in this time period are Byron Dorgan, Dale Bumpers, Thomas Harkin, Christopher Murphy, and Paul Wellstone. The same analysis for the Republican party shows that TV-TBIP identifies Jesse Helms, a Senator from North Carolina, as the most conservative Republican. This is in line with how this Senator is depicted in the media; for example, the New York Times (see Holmes 2008) stated that Helms was “bitterly opposed” to federal financing for research and treatment of AIDS that he believed was God’s punishment for homosexuals (see, e.g., Noden 2007). According to the DW-Nominate score, Jesse Helms is also ranked as the most conservative Republican for, for example, sessions 106 and 107.

3.2 | Temporal Change, Drivers of Partisanship, and Partisan Phrases

GST requires manual specification of the topics based on domain knowledge. TV-TBIP facilitates the inference of topics in a data-driven manner, thereby enabling an extension where topical content can change over time.

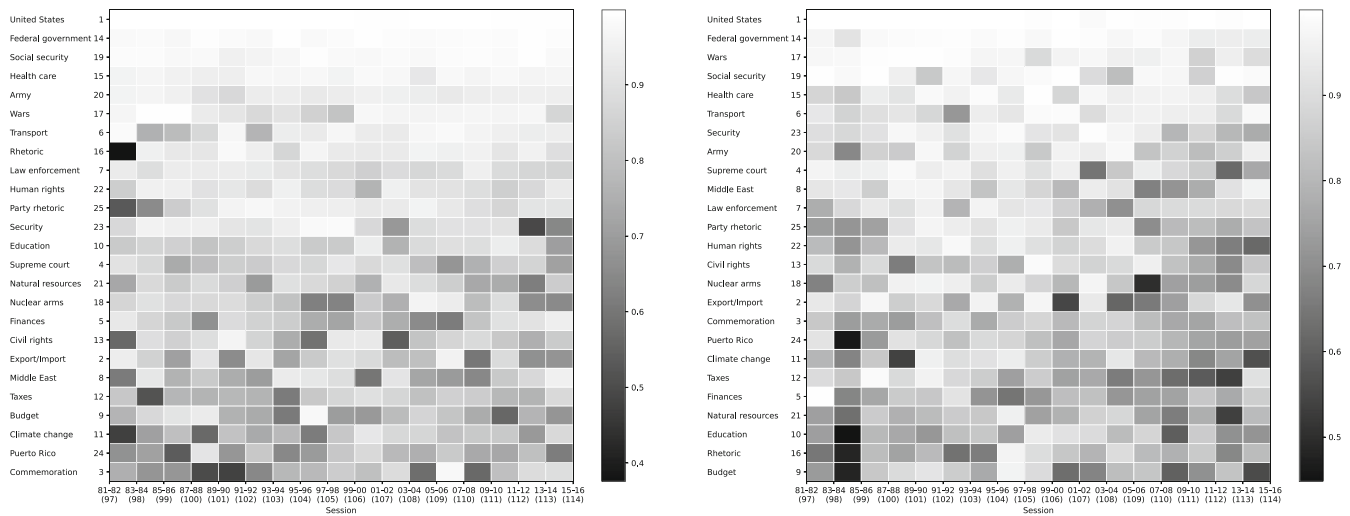


FIGURE 3 | Left: Similarity of the topics for consecutive sessions based on the term compositions of the neutral topics. Darker values indicate a higher change in term composition. Right: Similarities between the term compositions of the positive and negative ideological topics for each session. Darker values indicate a higher polarization for that topic in a specific session. The most frequent terms associated with the each of the topics can be found in Table 1.

taxes in general, but moves from a discussion on *tax cuts* to focus more on taxation and the *middle class* from the first to the last session. Of specific interest is also Topic 11, which represents a climate change/public health topic. For this topic, *nuclear waste* is a prevalent term in session 97, which changes to *climate change* being a prevalent term in session 114. Figure 2 displays the evolution of that topic using word clouds across all sessions. It changed from the discussion of *nuclear waste* over *acid rain* to *climate change* and *public health* related issues.

To assess stability of topic composition in a systematic way, we determine the cosine similarity between the estimated term intensities of the same topic for two consecutive sessions. Figure 3 on the left displays a heat map of these cosine similarities. The topics are sorted by their mean cosine similarity with the most stable one being on the top and the most volatile one at the bottom. For this comparison, the term intensities of the neutral topics were used.

On the top, we find Topic 1 (the US topic), which is the most stable topic over time. Other topics with stable topical content over time are Topic 14, which is concerned with the federal and local governments, and Topic 19, which is about social security. Among the three most volatile topics we find again Topic 11 (see Figure 2). In addition, the largest change in term composition of topics occurs for Topic 23 from sessions 112 to 113 with still quite a substantial change in term composition to the subsequent session 114. Inspecting the most frequent terms for Topic 23 for sessions 112 and 113 indicates that the term *national security* is substituted by *homeland security* and even more important *immigration* issues started to be also raised in that topic.⁴

3.2.2 | Topics as Drivers of Partisanship

In the following, we analyze which topics are main drivers of partisanship. We create again a heat map of the cosine similarity matrix, this time comparing the term frequencies of positive and

negative topics for each topic and session. The term prevalences of the positive and negative topics are determined using an ideal point of 1 (i.e., liberal) and −1 (conservative). The results are displayed in Figure 3 on the right. Topics are again sorted by their mean cosine similarity with the least polarizing being on the top and the most polarizing one at the bottom.

We find that Topic 1 is the most “neutral,” that is, the topic where the differences between the terms used by liberal and conservative speakers are smallest. In addition, we also observe low discordance for Topic 14, which is about the federal government, and Topic 17, which is about war veterans. Topic 11 on climate change/public health is close to the bottom of the heat map in Figure 3 on the right, indicating that there is less congruence in the wording of Democrats and Republican when talking about these issues. The same holds for Topic 21, another energy topic that is ranked as the fourth least congruent topic. Only Topic 9 (which is about government shutdowns and budget control), Topic 16 (which is a rhetoric topic) and Topic 10 (which is about education) have a lower concordance.

3.2.3 | Partisan Phrases

GST present estimated partisan phrases for selected periods, that is, phrases that allow to differentiate most between a Republican and a Democrat speaker (see Table 1 on p. 1325, which provides the 10 most important partisan phrases for every 10th session). In the TV-TBIP model, partisan phrases emerge on the topic level because the term distributions of the topics are influenced by the polarity scores and the ideal points of the speakers. To identify partisan phrases on topic level, we inspect the term compositions of the topics for a liberal speaker with ideal point value 1 and a conservative one with ideal point value −1. More formally, we compare the term distribution $\beta \exp\{\eta\}$ with $\beta \exp\{-\eta\}$. Table 2 displays the most frequent terms of each topic for these speakers.

TABLE 2 | Most frequent positive (left) and negative (right) terms for each topic for session 114.

#	Topic	Most frequent positive terms	Most frequent negative terms
1	United States	united states, puerto rico, trade agreements, (0.416) (0.04) (0.01) transpacific partnership, trade promotion (0.009) (0.007)	united states, trade agreements, (0.277) (0.034) promotion authority, trade promotion, (0.021) (0.021) currency manipulation (0.015)
2	Export/import	exportimport bank, small businesses, (0.035) (0.03) american workers, minimum wage, exim bank (0.029) (0.029) (0.026)	small businesses, economic growth, (0.055) (0.026) american workers, exportimport bank, (0.017) (0.013) labor relations (0.011)
3	Commemoration	medical research, national institutes, (0.039) (0.019) state university, passed away, high school (0.017) (0.015) (0.013)	high school, pay tribute, state university, (0.019) (0.018) (0.017) passed away, public servant (0.016) (0.012)
4	Supreme court	supreme court, attorney general, district court, (0.122) (0.034) (0.016) court nominee, judge garland (0.015) (0.015)	supreme court, justice scalia, attorney general, (0.13) (0.028) (0.021) presidential election, confirmation process (0.016) (0.011)
5	Finances	wall street, federal reserve, financial crisis, (0.085) (0.015) (0.012) financial institutions, consumer protection (0.009) (0.008)	wall street, federal reserve, protection act, (0.029) (0.018) (0.016) banking housing, monetary policy (0.012) (0.01)
6	Transport	home state, important issue, (0.075) (0.015) critically important, transportation system, (0.015) (0.013) roads highways (0.011)	home state, important issue, (0.073) (0.023) critically important, faa reauthorization, (0.014) (0.011) bottom line (0.009)
7	Law enforcement	law enforcement, gun violence, (0.067) (0.024) background checks, mental health, (0.022) (0.021) background check (0.021)	law enforcement, human trafficking, (0.078) (0.035) mental health, criminal justice, sex trafficking (0.017) (0.016) (0.012)
8	Middle East	national security, middle east, united states, (0.139) (0.028) (0.023) al qaeda, civil war (0.013) (0.01)	national security, united states, middle east, (0.113) (0.034) (0.033) islamic state, al qaeda (0.018) (0.016)
9	Budget	government shutdown, control act, (0.02) (0.02) appropriations process, budget deal, (0.015) (0.014) budget caps (0.012)	balanced budget, national debt, (0.029) (0.022) appropriations process, budget act, control act (0.017) (0.014) (0.011)
10	Education	high school, left behind, higher education, (0.058) (0.024) (0.022) child left, student loan (0.02) (0.018)	high school, left behind, child left, (0.044) (0.029) (0.024) health education, education labor (0.019) (0.019)
11	Climate change	climate change, public health, fossil fuel, (0.117) (0.031) (0.029) carbon pollution, global warming (0.023) (0.018)	public health, clean water, climate change, (0.044) (0.044) (0.034) zika virus, water act (0.024) (0.022)
12	Taxes	middle class, republican budget, tax credit, (0.062) (0.034) (0.023) tax breaks, tax code (0.021) (0.012)	tax code, middle class, tax reform, tax credit, (0.036) (0.035) (0.03) (0.024) income tax (0.01)
13	Civil rights	civil rights, voting rights, rights act, (0.04) (0.027) (0.022) citizens united, equal pay (0.02) (0.019)	civil rights, equal pay, fair housing, (0.037) (0.017) (0.014) voting rights, rights act (0.012) (0.011)
14	Federal government	federal government, private sector, (0.122) (0.036) taxpayer dollars, federal agencies, (0.019) (0.019) local governments (0.012)	federal government, private sector, (0.146) (0.028) taxpayer dollars, federal agencies, waste fraud (0.026) (0.015) (0.015)

(Continues)

TABLE 2 | (Continued)

#	Topic	Most frequent positive terms	Most frequent negative terms
15	Health care	health care, affordable care, care act, (0.122) (0.055) (0.052) health insurance, planned parenthood (0.045) (0.043)	health care, planned parenthood, (0.167) (0.028) health insurance, affordable care, care act (0.025) (0.024) (0.023)
16	Rhetoric	next week, equally divided, en bloc, (0.044) (0.02) (0.016) important legislation, final passage (0.012) (0.012)	equally divided, en bloc, important legislation, (0.062) (0.049) (0.032) next week, final passage (0.029) (0.016)
17	Wars	puerto rico, world war, war ii, vietnam war, (0.049) (0.047) (0.036) (0.016) loved ones (0.013)	world war, war ii, puerto rico, choice act, (0.053) (0.049) (0.03) (0.017) veterans administration (0.013)
18	Nuclear arms	nuclear weapon, united states, (0.039) (0.033) nuclear weapons, nuclear program, (0.033) (0.023) irans nuclear (0.018)	united states, nuclear weapons, (0.036) (0.03) nuclear program, middle east, nuclear weapon (0.023) (0.02) (0.02)
19	Social security	social security, trust fund, highway trust, (0.115) (0.042) (0.022) older americans, security benefits (0.014) (0.009)	social security, trust fund, highway trust, (0.12) (0.051) (0.017) security administration, disability insurance (0.012) (0.009)
20	Army	air force, defense authorization, (0.037) (0.036) armed services, national defense, (0.033) (0.03) national guard (0.018)	air force, armed services, national defense, (0.055) (0.054) (0.045) defense authorization, national guard (0.043) (0.024)
21	Natural resources	energy efficiency, tar sands, (0.026) (0.023) water conservation, conservation fund, (0.022) (0.02) clean energy (0.018)	keystone xl, natural gas, natural resources, (0.024) (0.021) (0.02) xl pipeline, energy security (0.02) (0.016)
22	Human rights	human rights, state department, (0.065) (0.036) foreign relations, united nations, (0.028) (0.02) american citizens (0.017)	religious freedom, human rights, (0.047) (0.037) state department, foreign relations, (0.033) (0.027) religious liberty (0.022)
23	Security	homeland security, cyber security, (0.109) (0.027) usa freedom, freedom act, immigration reform (0.019) (0.018) (0.018)	homeland security, cyber security, (0.082) (0.034) inspector general, freedom act, usa freedom (0.025) (0.012) (0.012)
24	Puerto Rico	first responders, local communities, (0.013) (0.013) economic development, emergency funding, (0.011) (0.011) grant program (0.009)	puerto rico, economic development, (0.019) (0.016) foster care, local communities, much needed (0.014) (0.012) (0.01)
25	Party rhetoric	republican colleagues, bipartisan basis, (0.019) (0.015) republican leadership, good friend, (0.012) (0.012) put together (0.011)	democratic colleagues, bipartisan basis, (0.018) (0.015) bipartisan way, good friend, democratic friends (0.015) (0.013) (0.011)

Note: Values in parenthesis denote appearance rates.

In this analysis, our focus is on session 114, which occurs at the end of the observation period and where the latent dimension distinctly enables discrimination between Republican and Democrat speakers. In line with GST, who identify *puerto rico* as a partisan phrase, we also see that *puerto rico* is among the most frequent positive or negative terms for Topics 1, 17, and 24. For example, for Topic 17, we estimate an appearance rate of 0.049 for Democrats and 0.03 for Republicans. GST find rates of 42 (R) and 79 (D) per 100,000 phrases. Further, also in line with GST, *religious freedom* appears among the most frequent term only for Republicans. GST find this term to appear 34 times per 100,000 phrases for Republicans and 4 times for Democrats. The TV-TBIP

results also highlight the topic where this word appears, that is, Topic 22.

In addition to GST, for example, our results indicate that for Topic 21—which is about natural resources and energy—a speaker with a positive, that is, a liberal, ideal point uses terms like *energy efficiency* or *clean energy* when talking about this topic while a conservative one uses terms like *keystone xl* (a pipeline project by TC Energy) or *energy security*. For Topic 5, which is about monetary policy, we find on the liberal side terms like *financial crisis* and *consumer protection* but on the conservative side *banking housing* and *monetary policy*. For Topic 11, which is

about climate change/public health, the term *climate change* is used three times more often by a speaker with ideal point 1 compared to one with -1 . GST also detect climate change as a partisan phrase with 94 (D) and 23 (R) appearances per 100,000 phrases.

4 | Conclusion and Discussion

This study replicates GST in a wide sense through an alternative modeling approach, analyzing US Senate speeches from 1981 to 2017. The differences of our approach compared to GST are threefold: First, TV-TBIP combines the class of topic models with ideal point models. Thus, researchers do not have to manually specify topics a priori using key terms. Second, our approach is unsupervised. This implies it can also be used to analyze data for which no information about author positions is known hitherto as pre-labeling is not required (party membership). Third, our model is able to detect the accordance of political parties on topic level through polarity- and topic-specific term distributions in a time-dynamic way.

Our results highlight a sharp rise in partisanship during the 1990s, which aligns with GST's conclusions. By inspecting individual ideal points, we compare Senators' positions on a latent scale. We also observe shifts in topic content over time. The climate change/public health topic emerges as a significant driver of partisanship, showing high term discordance between parties, while the natural resources and energy topic also differentiates between speakers across party lines. In contrast, some topics, like war veterans, display similar term compositions across both parties.

This sets the scene for further research which could include, for example, alternative ways of modeling time variation in this model framework (see Vavra et al. 2024a) or the incorporation of speaker-specific covariates that influence ideological positions (see Vavra et al. 2024b).

Acknowledgments

Open Access funding provided by Universitat Salzburg/KEMÖ. [Correction added on 10 May 2025, after first online publication: KEMÖ-DEAL funding statement has been added.]

Data Availability Statement

The data that support the findings of this study are openly available in Stanford Libraries [distributor] at https://data.stanford.edu/congress_text.

Open Research Badges



This article has been awarded Open Data Badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. Data is available at <https://doi.org/10.15456/jae.2025027.1657744625>.

Endnotes

¹ GST use hein-daily and hein-bound transcripts (see https://data.stanford.edu/congress_text), where hein-bound covers the time period 1873–2011 and hein-daily 1981–2017. Given the focus on partisanship

during the 1990s and later in GST, we make only use of the hein-daily speeches covering 1981–2017 in our analysis.

² This modeling approach has also been pursued to study central bank communication in the euro area (see Feldkircher et al. 2024).

³ Indeed, we observe a similarly high value for all sessions.

⁴ Note that similar observations have also been discussed in <https://www.everysreport.com>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.